To start off, we will look at the core ideas of the data mesh, as well as the benefits and challenges associated with it. We'll also give you our definition of a data mesh, focused on outcomes and practicality.

1.1 Data mesh 101

The data mesh paradigm in this book is all about decentralizing responsibility. For instance, say the development team for the customer registration component of a company also creates a dataset to analyze registered customers. The team ensures that this dataset is in an easy-to-digest format by transforming the data (e.g., to a CSV file) and serving it the way the consumers would like it (e.g., on a central file-sharing system).

But this deceptively simple definition has a lot of implications because, in most companies, data is handled as a *by-product*. It is usually turned into value only after being put as a by-product into storage, then pulled into a central technology by a centralized data team, and then picked up again by decentralized actors. This actor may be an analyst in the marketing department, a recommendation system used in a marketing campaign, or a frontend display. Figure 1.1 depicts a common form of data architecture, both organizational and technical. We have also done our best to show its pitfalls.